Hanisha Pulimaddi

Software Engineering · Machine Learning & AI Systems · Product, Growth and Consultancy
hani.pulimaddi@outlook.com | +61 451 619 712 | Sydney, NSW, Australia
LinkedIn | GitHub | Portfolio

SUMMARY

Software engineering student who builds and deploys systems end to end rather than stopping at the notebook, from model training and cloud infrastructure through to customer discovery and pitching. Custom-trained a YOLOv5 detector, provisioned a containerised ML service on AWS ECS Fargate, and architected LLM pipelines where a deterministic layer owns the calculation. As technical founder in a startup accelerator residency, ran 200+ outreach contacts and 5+ discovery interviews, delivered 10+ pitches, and repositioned the product when the interviews contradicted the original thesis. Looking to bridge my technical knowledge with my interest in startup, consultancy and growth roles.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, C

Machine Learning: scikit-learn, logistic regression, random forests, class imbalance handling, stratified cross-validation, ROC-AUC / F1 / AUPRC, threshold tuning, probability calibration

Deep Learning & CV: PyTorch, YOLOv5 (custom-trained), DenseNet121 / torchxrayvision, inference serving

LLM & AI Systems: Claude API, Groq, retrieval-augmented pipeline design, JSON-schema extraction contracts, citation-traced extraction

Backend & Data: FastAPI, SQLAlchemy, REST APIs, JWT authentication, Gunicorn, Uvicorn, PostgreSQL, RDS, SQLite

Frontend: React 19, TypeScript, Leaflet

Cloud & DevOps: AWS (ECS Fargate, ECR, ALB, VPC, RDS, S3, Secrets Manager, IAM, CloudFront, Route 53, ACM), Terraform, Docker, GitHub Actions (OIDC), CI/CD

Practices: Git feature branching, infrastructure as code, zero-downtime deployment, database migrations, least-privilege IAM, integration debugging, cost and performance profiling

Product & Growth: customer discovery and user interviews, problem framing, PRD and requirements writing, wireframing and annotated screens, prototyping (Base44), cold outreach at scale, pitching to investor and mentor audiences, cost modelling and unit economics, stakeholder management

EXPERIENCE

Founder in Residence | Arrayah (Chapter 3 Residency), Drummoyne, NSW *Apr - Jul 2026*

- Technical founder of an automated compliance platform for architects, owning product direction, architecture and build across a 4-month residency.
- Ran outreach to 200+ professionals on LinkedIn and email, converting 5+ into discovery interviews that reset the product thesis.
- Repositioned the product from rule extraction to rule interpretation after architects made clear that surfacing a clause did not answer their question.
- Delivered 10+ pitch iterations, defending technical trade-offs to non-technical evaluators.

Technical Consultant | Lou's Place (non-profit, one-week engagement), Redfern, NSW *Sep 2026*

- Ran discovery with frontline caseworkers at a crisis centre serving 818 women a year, tracing their abandoned services directory to a maintenance burden nobody owned.
- Scoped and delivered a caseworker referral tool inside the one-week engagement, designing capture so directory updates fall out of work caseworkers already do, not added upkeep.
- Cost-modelled the build to under $2/month, keeping it viable for an org with no recurrent funding.

Co-founder | CuzYNot, Sydney, NSW *Sep 2026 - Present*

- Co-founded a try-everything community club that takes on a new skill every week (trails, tennis, salsa, latte art), built on permission to be a beginner in public.
- Designed the format for zero-barrier entry, no experience required, solo attendance encouraged, which converted 20+ registrations.
- Moved registrants into a WhatsApp community ahead of the event rather than leaving them on the ticketing platform, retaining 80% for the next session.

Student Guide | UTS Library, Sydney, NSW *Feb 2026 - Present*

- Process 200+ items weekly across shelving, restocking and holds, keeping requested material available.
- Assisted 100+ new students at orientation with library and campus resources, easing their first week.
- Support front-line operations across service points alongside library assistants and management staff.

PROJECTS

OmniScan — Medical Imaging Platform on AWS *Aug 2026 - Present*

- Serving chest X-ray inference from a pretrained DenseNet121 classifier, with a Groq-hosted LLM converting per-pathology scores into a structured radiology-style report.
- Provisioning the full AWS stack in Terraform (ECS Fargate, ALB, two-AZ VPC, RDS PostgreSQL, S3, Secrets Manager) with remote state and locking, so the environment rebuilds from code rather than console clicks.
- Cut the Docker image from 2.5 GB to 576 MB (77%) with multi-stage builds after profiling showed the runtime needed none of the build layers, and right-sized ECS task compute from measured profiling.
- Automated releases through GitHub Actions over OIDC, removing stored AWS keys entirely; health checks and a separate pre-deploy migration step keep deploys zero-downtime.

WayFind — Agentic Crisis Centre Referral Tool | Hack for Humanity Hackathon *Sep 2026*

- Designed a 3-tier agentic architecture separating query-time verification, referral-time capture and aggregate insight, scoping model autonomy to 1 of 5 pipeline stages and leaving eligibility and ordering to deterministic code.
- Built a 15-second voice debrief on the Web Speech API behind a JSON-schema extraction contract, so free-form speech landed as validated structured records and the directory rebuilds itself from caseworker notes.
- Scoped the system to store no client identifiers, keeping it usable where confidentiality is non-negotiable.
- Shipped a working Claude-backed build plus a PRD, 8 annotated screens, a wireframe set and 5 self-contained HTML prototypes inside a 2-day window.

Automated Compliance Checker — LLM Rule Extraction | Arrayah Chapter 3 Residency *Apr - Jul 2026*

- Built an end-to-end compliance pipeline (FastAPI backend, React and Leaflet frontend, LLM rule extraction with clause-level citations, geometric envelope engine) and shipped it through CI/CD.
- Validated rule extraction against 26 verified NSW Planning API addresses at 100% recall and precision, setting the accuracy baseline every later change was tested against.
- Split the architecture so the LLM owns extraction and a deterministic geometric engine owns the compliance calculation, containing model error to a stage users could check.
- Rebuilt the system from rule extraction to rule interpretation after 15+ architect interviews.

FridgeIQ — Object Detection & Full-Stack Delivery | Deep Learning & CNN, UTS *Feb - May 2026*

- Custom-trained a YOLOv5 detector on fridge imagery and served it behind a FastAPI inference endpoint (SQLAlchemy over SQLite, JWT auth), reducing inventory entry to one photo upload.
- Shipped the full path from model to user, with a React and TypeScript client generating priced grocery lists.
- Traced and fixed 80+ integration bugs across model serving, database queries and authentication, which taught me to reproduce failures at the API layer before touching frontend code.
- Kept 20+ incremental commits on feature branches, a history granular enough to isolate and roll back regressions.

Default Prediction System — BNPL Credit Risk | Machine Learning, UTS *Sep 2025 - Jan 2026*

- Trained and benchmarked 3 models over 255K+ loan records under 3-fold stratified cross-validation, selecting on AUPRC and F1 because accuracy carried no signal at the base rate.
- Tuned decision thresholds and calibration curves so output arrived as risk scores a credit team could act on directly.
- Quantified the 65–70% recall operating point as an approve/decline cost trade-off between missed defaults and rejected good customers, making the threshold a business choice rather than a default.

Deloitte Technology & Transformation Case Competition | ClinIQ Advisory *Aug 2026*

- Designed and pitched In Sync to the AI and Data Team at Deloitte Sydney Headquarters, covering product strategy, system architecture, data governance and operational feasibility.
- Architected a hybrid edge-cloud platform (PostgreSQL, Kafka, HL7 v2/FHIR, event-sourced object storage), modelling a 45% turnaround-time reduction.
- Designed a privacy-preserving data model with tokenisation at the network edge, satisfying data and access-control requirements.

LEADERSHIP & INVOLVEMENT

- Gender Equity Ambassador, Women in Engineering & IT, UTS (Feb 2025 - Aug 2026), ran 3+ STEM events reaching 200+ participants and mentored 50+ students.
- Executive Member, UTS Hiking Club (Aug 2025 - Aug 2026) and Volunteer, UTS Peer Network (Jan - Aug 2025), event logistics and peer support across 200+ students.
- Writing - four published essays, including a referenced piece on semiconductor supply chains and geopolitics (Sep 2026).

EDUCATION

University of Technology Sydney | Bachelor of Engineering (Honours), Software Engineering *Feb 2024 - Dec 2027*

Sancta Maria International School | ICE (A-Level & AS-Level), IGCSE, Telangana, India *Jun 2015 - Apr 2023*